Open-World Explained-Rarity Filtering for Health-Insurance Claim Fraud: Problem Formulation, Benchmark Design and Preliminary Synthetic Evidence

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***Abstract*—Health-insurance fraud detectors trained on known fraud categories can force novel schemes into familiar classes and confuse clinically rare but legitimate claims with fraud. We formulate *explained-rarity filtering*: a claim is escalated only when it is unfamiliar and unexplained by clinical and provider evidence. We report three distinct synthetic experiments: an earlier tabular prototype; a standalone, independently trained LSTM, 1D-CNN and Transformer comparison over 36 model/fold/seed fits; and a temporal relation-summary pilot that implements additional OW-ERF stages. In the standalone comparison, the Transformer achieved mean known-class accuracy 0.809 and mean novel-mechanism AUROC 0.693 across the four held-out-mechanism folds and three seeds, while results were highly variable across folds. These experiments are not a single integrated HGT/R-GCN system and do not use real insurance claims. We report the methods, measured outcomes and remaining limitations without claiming real-world performance.**

***Index Terms*—health-insurance fraud, open-world recognition, conformal prediction, anomaly detection, graph neural networks**

I. Introduction

Large claim volumes in health-insurance systems, including IRDAI-regulated insurance and the AB-PMJAY scheme in India, make exhaustive manual review impractical. Fraud mechanisms such as upcoding, unbundling, phantom billing and collusive referral rings also change over time. Three problems follow.

1) **Closed-world classification.** A softmax classifier assigns a new scheme to the nearest known class, often with high confidence.
2) **Rarity is not fraud.** Pure anomaly scores flag legitimate severe cases because they are rare, complex and expensive.
3) **Alert fatigue.** Raw scores carry no statistical meaning, so a fixed threshold drifts and review budgets are exceeded.

Contributions. (1) A problem formulation, *explained-rarity filtering*, that separates unfamiliarity from unexplainedness. (2) A pipeline (OW-ERF) combining post-hoc novelty fusion, split-conformal rejection and a heteroscedastic clinical expectation filter. (3) A benchmark protocol with a synthetic claim generator, leave-one-mechanism-out evaluation and leakage control. (4) Preliminary evidence from a tabular prototype, including where the filter hurts.

II. Related Work

Open-set and open-world recognition. Open-set recognition [1], [2] and out-of-distribution (OOD) baselines [3]–[5] score unfamiliarity from logits or embeddings. On graphs, OpenWGL [6] rejects unseen-class nodes with a variational graph autoencoder, and GNNSafe [7] applies energy-based detection to graph neural networks.

Graph fraud and anomaly detection. CARE-GNN [8] counters camouflage with reinforcement-learned neighbor selection, and BWGNN [9] addresses heterophily. DOMINANT [10] and the PyGOD library [11] cover reconstruction-based detection, while BOND [12] and GADBench [13] provide benchmarks. These methods output novelty or anomaly scores and do not separate rare-but-clinically-explained claims from unexplained ones.

Conformal prediction. Split-conformal methods [14]–[16] give finite-sample marginal guarantees under exchangeability. Barber et al. [17] study relaxations, and Huang et al. [18] apply conformal prediction to graph neural networks. Temporal claim streams violate exchangeability, so our guarantee is approximate and is tested empirically.

Heteroscedastic regression. Predicting a mean and a variance under a Gaussian likelihood [19], [20] lets the expected spread grow with severity.

Healthcare fraud. Prior work applies supervised and unsupervised learning to Medicare-style data [21].

III. Problem Formulation

Let a claim v have features x_v, a graph neighborhood, and a label y_v from benign, known mechanisms, or novel mecha-

nisms. Training sees only benign and known mechanisms. At test time a claim is *unfamiliar* if its conformal p-value satisfies $p_v \le \alpha$, and *unexplained* if its clinical inconsistency ratio a_v exceeds a calibrated threshold τ. A claim is escalated if and only if it is both.

To prevent temporal leakage, the embedding of claim v at adjudication time t uses only the causal neighborhood

$$\mathcal{N}_r(v;t) = \{u : (u,v) \in E_r,\ t(u,v) \le t\}. \tag{1}$$

IV. Method

Fig. 1 shows the pipeline.

A. Encoder

The target architecture specifies a timestamped heterogeneous graph of claims, beneficiaries, hospitals, doctors, admissions, packages and diagnoses, with time-aware edges and an R-GCN or HGT encoder using PyTorch Geometric [22]–[24]; the full arbitrary-node PyG encoder is not yet implemented. Three complementary branches are separately implemented and trained in the released synthetic benchmark: an LSTM over recent claim-history sequences, a 1D-CNN over ordered procedure-code tokens, and a Transformer over relational-context tokens. The standalone experiment compares each architecture independently rather than claiming that it is already fused with a full HGT/R-GCN. The three models use different input modalities by design, so the comparison is a comparison of architecture-plus-modality pipelines. A separate relation-summary pilot implements causal relation-specific attention and additional OW-ERF stages, but it is not a full arbitrary-node heterogeneous graph encoder.

B. Novelty score

From classifier logits z, the energy is

$$E(x) = -T \log \sum_k \exp(z_k/T). \tag{2}$$

The Mahalanobis score $M(h)$ is the minimum over known classes of the squared Mahalanobis distance of embedding h to the class mean under the class covariance. With F_E and F_M the empirical CDFs of E and M on training data, the fused score is

$$u = \tfrac{1}{2}F_E(E) + \tfrac{1}{2}F_M(M). \tag{3}$$

C. Conformal rejection

On a calibration set disjoint from training and containing only known-class claims, with n calibration scores u_i, the p-value is

$$p_v = \frac{1 + \#\{i : u_i \ge u_v\}}{n+1}, \tag{4}$$

and a claim is rejected as unfamiliar when $p_v \le \alpha$ with $\alpha = 0.05$.

D. Explained-Rarity Filter

A heteroscedastic model trained on benign claims only predicts a mean μ_j and standard deviation σ_j for $j \in \{\text{log-amount}, \text{length of stay}\}$ from *exogenous* inputs (a severity proxy from prior history, comorbidity count, age) by minimizing

$$\mathcal{L}_g = \frac{1}{N}\sum_i \sum_j \Big[\ln \sigma_{ij} + \frac{(o_{ij}-\mu_{ij})^2}{2\sigma_{ij}^2}\Big]. \tag{5}$$

The inconsistency ratio is

$$a_v = \max_j \max\Big\{0, \frac{o_j - \mu_j}{\sigma_j}\Big\}. \tag{6}$$

The threshold τ is set as a quantile of a_v on benign calibration claims rather than by hand. The expectation model must not condition on fields a fraudster controls, such as the claimed diagnosis code; otherwise upcoding would make fraud look explained.

E. Additional OW-ERF pilot components

The separate relation-summary pilot implements episodic pseudo-novel energy-margin training, energy/Mahalanobis fusion, split-conformal rejection, legitimate-only clinical expectation and structural-reference thresholds, explained-rarity filtering, HDBSCAN grouping (with DBSCAN fallback when HDBSCAN is unavailable), cluster cards and audit-oriented metrics. It uses past-only entity summaries over six relation types, not a full PyG HGT/R-GCN over arbitrary typed nodes. The standalone LSTM/CNN/Transformer experiment is a separate benchmark and must not be reported as the full graph model.

V. Experimental Protocol

Data. A synthetic generator produces benign claims, 5% rare-severe legitimate claims, and four mechanisms: upcoding, unbundling, phantom billing, and collusion (represented in the prototype by two proxy features only). The fraud rate is 10%.

Splits and metrics. Three experiments are reported separately. The earlier tabular prototype uses five random seeds and its own i.i.d. split. The standalone deep comparison uses three generator seeds (42, 43, 44), four leave-one-mechanism-out folds, an internal known-only validation subset, a separate known-only calibration split, and an untouched test split, yielding 36 independent model fits. Known-class accuracy is calculated only on test claims from classes observed during training; novel-mechanism AUROC/AP treat the held-out mechanism as positive for novelty scoring. The v0.2 relation-summary pilot uses chronological train, selection, calibration and deployment windows. These results are synthetic, and the standalone benchmark does not establish real-world performance.

Baseline. Isolation Forest [25], with its alert count matched to the number of OW-ERF escalations.

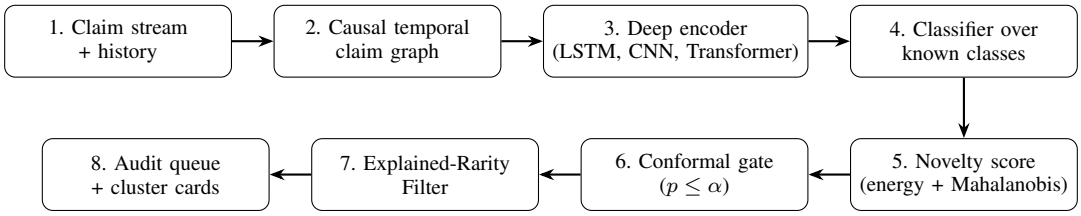


Fig. 1. Pipeline of OW-ERF. Stages 1–4 learn representations; stages 5–8 make the decision.

TABLE I
LEAVE-ONE-MECHANISM-OUT RESULTS (SYNTHETIC DATA, TABULAR PROTOTYPE, MEAN OF 5 SEEDS). RECALL IS ON THE HELD-OUT MECHANISM; "RARE FLAG" IS THE FRACTION OF RARE-SEVERE LEGITIMATE CLAIMS FLAGGED.

Held-out mechanism	Recall conf.	Recall +ERF	Rare flag conf.	Rare flag +ERF
Upcoding	0.574	0.560	0.026	0.006
Unbundling	0.493	0.372	0.020	0.003
Phantom billing	0.496	0.411	0.133	0.026
Collusion (proxy)	0.586	0.211	0.016	0.002

VI. PRELIMINARY RESULTS

Table I retains the earlier tabular-only reference results (means over five seeds). The conformal stage rejects between 4.3% and 4.6% of benign claims, close to the nominal 5% on this i.i.d. synthetic split; this does not demonstrate validity under temporal drift. After the filter, benign escalation falls to about 0.5%. The filter cuts flags on rare-severe legitimate claims most clearly for phantom billing ($13.3\% \rightarrow 2.6\%$). It also lowers recall: for collusion the drop is large ($0.586 \rightarrow 0.211$), because those claims look clinically plausible and would require graph structure that this prototype lacks. At a matched alert count, Isolation Forest recalls only 0.02–0.04 of the three attribute-level mechanisms (0.20 for collusion) while flagging 22%–28% of rare-severe legitimate claims. All of these results are conditional on the generator.

A. Standalone deep-model comparison

Table II reports the mean and standard deviation across the 12 model/fold/seed runs per architecture (four held-out mechanisms by three generator seeds). Known-class accuracy excludes the held-out mechanism; novelty metrics treat the held-out mechanism as novel. This distinction is important because the model may classify known classes reasonably while failing to reject unfamiliar fraud.

The Transformer has the strongest mean known-class accuracy and novel AUROC in this 5-epoch pilot, but the high dispersion and low mean novel recall show that performance is not yet robust. The average known false-rejection rates at $\alpha = 0.05$ were 0.050 (LSTM), 0.047 (1D-CNN), and 0.045 (Transformer); this calibration target applies to known classes, not unseen fraud. The full JSON/CSV and selected checkpoints are generated by the dedicated GitHub Actions workflow. These metrics are synthetic generator results, not real insurance claim results.

B. Controlled architecture comparison with shared inputs

The native standalone comparison above changes both architecture and input modality. To isolate the architecture effect more fairly, we ran a paired controlled comparison: the LSTM, a continuous-input 1D-CNN, and Transformer received the same standardized six-token, 16-channel relational-context tensor and the same padding mask. Scaling was fitted only on valid training tokens; padded positions were zeroed, the LSTM packed valid sequence lengths, the CNN used mask-aware pooling, and the Transformer used its key-padding mask. The experiment used the same four held-out mechanisms and generator seeds 42, 43, and 44, for 36 model fits per mode (72 total). Checkpoints were selected using validation cross-entropy; the test split was not used for model selection. The 25-epoch pilot is preserved in a separate report; Table III presents the completed 60-epoch follow-up.

The large LSTM and CNN known-class-accuracy changes under shared inputs show that the original score gap was substantially confounded by modality: collusion information is encoded in history, unbundling in code tokens, and upcoding/phantom cues in the claim-context view. The Transformer is numerically unchanged because its native input was already that context view. Under shared context, novelty AUROC improves for LSTM and CNN, but dispersion remains high and novel recall at the conformal threshold is still low/variable. The shared-context known false-rejection rates were 0.048 ± 0.019 (LSTM), 0.054 ± 0.015 (1D-CNN), and 0.061 ± 0.021 (Transformer); the Transformer remains slightly above the nominal 5% rate. Validation-selected best epochs reached the 60-epoch cap in 3/12 shared-context LSTM runs and 1/12 shared-context CNN runs (none of the Transformer runs), so some LSTM/CNN folds may still benefit from a larger epoch budget or early-stopping study. With three generator seeds, these descriptive metrics do not establish a statistically decisive architecture ranking. All results are synthetic and do not establish performance on real insurance claims. Exact reports and checkpoints are committed under `results/controlled_deep_model_comparison/controlle` the 25-epoch pilot remains separately archived.

VII. LIMITATIONS

All measured experiments reported here use synthetic data. The independent LSTM/CNN/Transformer comparison

TABLE II
INDEPENDENT DEEP-MODEL RESULTS ON THE SYNTHETIC HELD-OUT-MECHANISM BENCHMARK (MEAN ± STANDARD DEVIATION ACROSS 12 FOLDS/SEEDS PER MODEL).

Model	Known accuracy	Known macro-F1	Novel AUROC	Novel AP	Novel recall ($\alpha = 0.05$)
LSTM	0.630 ± 0.183	0.380 ± 0.105	0.576 ± 0.189	0.139 ± 0.277	0.156 ± 0.288
1D-CNN	0.565 ± 0.114	0.279 ± 0.048	0.561 ± 0.126	0.092 ± 0.093	0.120 ± 0.159
Transformer	0.809 ± 0.115	0.646 ± 0.137	0.693 ± 0.142	0.142 ± 0.215	0.156 ± 0.251

TABLE III
CONTROLLED DEEP-MODEL COMPARISON ON SYNTHETIC DATA (60-EPOCH MAXIMUM; MEAN ± STANDARD DEVIATION ACROSS 12 FOLD/SEED RUNS PER MODEL). NATIVE MODE USES EACH MODEL'S INTENDED INPUT; SHARED-CONTEXT MODE GIVES ALL MODELS THE SAME STANDARDIZED TOKENS.

Model	Native acc.	Shared acc.	Native AUROC	Shared AUROC	Shared AP	Shared novel recall ($\alpha = 0.05$)
LSTM	0.519 ± 0.098	0.979 ± 0.010	0.539 ± 0.130	0.721 ± 0.210	0.169 ± 0.105	0.250 ± 0.166
1D-CNN	0.896 ± 0.016	0.973 ± 0.016	0.546 ± 0.136	0.759 ± 0.171	0.188 ± 0.141	0.354 ± 0.325
Transformer	0.983 ± 0.009	0.983 ± 0.009	0.816 ± 0.090	0.816 ± 0.090	0.174 ± 0.103	0.349 ± 0.228

reaches only moderate held-out-mechanism AUROC on average and shows high fold-to-fold variance; the 5% conformal gate also has low recall for novel claims in this initial run. Known-class accuracy and novel-mechanism AUROC are different tasks and must not be collapsed into one accuracy claim. The standalone sequence models use generated proxies, and they are not integrated with a full graph encoder. The v0.2 relational pilot uses causal summaries of prior entity events, not an arbitrary-node PyG HGT/R-GCN; its package rates are parametric rather than curated from the official HBP 2.2 annex, and it does not ingest Synthea trajectories. The conformal guarantee is marginal and assumes exchangeability, which temporal streams violate, and it does not cover the explained-rarity filter. If severity evidence can be manipulated, fraud can be explained away. Only the declared small baseline set is currently run; full model baselines, ablations, repeated confirmatory runs and an audit of shortcut features remain necessary. Finally, wrongly escalated claims can delay care, so any real deployment requires human review and group-wise error analysis. The 25-epoch controlled architecture comparison indicates that modality substantially affects the native results; an extended 60-epoch controlled run is reported separately when complete.

VIII. CONCLUSION AND FUTURE WORK

The repository now contains an independently trained LSTM/CNN/Transformer synthetic benchmark, an earlier tabular OW-ERF prototype, and a temporal relation-summary pilot that implements additional open-world stages. The initial deep benchmark suggests that the Transformer is strongest on known-class classification, but novel-mechanism detection remains variable and weak at the configured conformal threshold. The next steps are to improve generator realism, complete Synthea/HBP grounding, implement a full temporal heterogeneous graph encoder, run the full baseline/ablation/seed plan, audit shortcuts and subgroup errors, and then evaluate on appropriately labelled real data.

DATA AND CODE AVAILABILITY

No patient data are used. The synthetic generator, tabular prototype, standalone deep-model benchmark, temporal relation-summary pilot, and policy-recommendation prototype are available at https://github.com/ishantpant2121/MUJ-DS-2430010379. The standalone comparison is reproducible from `scripts/train_evaluate_deep_models.py`; it writes JSON/CSV metrics and 36 model checkpoints to `results/standalone_deep_models/`. All deep and graph-style results use synthetic data; the separate public provider benchmark uses provider-level `PotentialFraud` labels and is not claim-level mechanism validation.

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